

Task-04

Date : 20/06/26

Understanding of Tokenization,
Embeddings, RNN and Transformers

Sentence 1

Sentence :

I am Lalitha Sowjanya

Tokenization :

["I", "am", "Lalitha", "Sowjanya"]

Embeddings :

I $\rightarrow$ [0.12, 0.34]

am $\rightarrow$ [0.81, 0.45]

Lalitha $\rightarrow$ [0.25, 0.11]

Sowjanya $\rightarrow$ [0.12, 0.45]

Sentence 2

Sentence :

The cat is sleeping

Tokenization :

["The", "cat", "is", "sleeping"]

Embeddings :

The $\rightarrow$ [0.70, 0.98]

cat $\rightarrow$ [0.11, 0.13]

is $\rightarrow$ [0.15, 0.23]

sleeping $\rightarrow$ [0.51, 0.32]

Sentence 3

sentence:

I am working on a project

Tokenization:

["I", "am", "working", "on", "a", "project"]

Embeddings:

I $\rightarrow$ [0.90, 0.78]

am $\rightarrow$ [0.01, 0.28]

working $\rightarrow$ [0.11, 0.23]

on $\rightarrow$ [0.55, 0.66]

a $\rightarrow$ [0.78, 0.99]

project $\rightarrow$ [0.01, 0.05]

Sentence 4

sentence:

My phone needs charge

Tokenization:

["my", "phone", "needs", "charge"]

Embeddings:

My $\rightarrow$ [0.56, 0.77]

phone $\rightarrow$ [0.98, 0.76]

needs $\rightarrow$ [0.89, 0.66]

charge $\rightarrow$ [0.99, 0.77]

Sentence 5

sentence :
Small language model

Tokenization :
["small", "language", "model"]

Embeddings:

small $\rightarrow$ [0.78, 0.66]

language $\rightarrow$ [0.98, 0.11]

model $\rightarrow$ [0.23, 0.33]

Sentence 6

sentence :
Company name is bighit

Tokenization:
["company", "name", "is", "bighit"]

Embeddings:

Company $\rightarrow$ [0.77, 0.83]

name $\rightarrow$ [0.11, 0.46]

is $\rightarrow$ [0.56, 0.69]

bighit $\rightarrow$ [0.73, 0.63]

Sentence 7

sentence :
Exams are coming

Tokenization:
["Exams", "are", "coming"]

Embedding:

Exams $\rightarrow$ [0.73, 0.69]

are $\rightarrow$ [0.32, 0.45]

coming $\rightarrow$ [0.66, 0.77]

Sentence 8

Sentence:

Multi PDF RAG chatbot

Tokenization:

["Multi", "PDF", "RAG", "chatbot"]

Embeddings:

Multi $\rightarrow [0.81, 0.64]$

PDF $\rightarrow [0.53, 0.74]$

RAG $\rightarrow [0.40, 0.81]$

chatbot $\rightarrow [0.22, 0.33]$

Sentence 9

Sentence:

Meeting is cancelled

Tokenization:

["Meeting", "is", "cancelled"]

Embeddings:

Meeting $\rightarrow [0.01, 0.02]$

is $\rightarrow [0.53, 0.62]$

cancelled $\rightarrow [0.22, 0.32]$

Sentence 10

Sentence:

I drank water

Tokenization:

["I", "drank", "water"]

Embeddings:

I $\rightarrow [0.73, 0.99]$

drank $\rightarrow [0.32, 0.41]$

water $\rightarrow [0.51, 0.67]$

RNN & Transformer

Sentence 1

My phone needs charge

RNN Workflow:

My → phone → needs → charge → output

- * RNN processes one word at a time
- * RNN may forget previously processed word if the sentence is too big.
- * May not know relations between words

Transformer Workflow:

My phone needs charge
 ↓ ↓ ✓
 Attention mechanism
 ↓
 output

- * Transformer process all words at the same time.

- * Attention mechanism identifies relationships between words

ex: My ↔ phone
 phone ↔ charge

- Understands relationships between words using attention mechanism

Sentence 2

Company name is bighit

RNN Workflow:

Company → name → is → bighit
output

* First reads "Company"

* Then it reads "name" while remembering "company"

* Next, "is" while remembering previous ones

* Finally, "bighit"

Transformer Workflow:

Company name is bighit

↓
Attention mechanism

↓
output

* Reads all words at a time

* Uses attention mechanism to understand grammar, context, relationships and meaning

ex: Company ↔ bighit
Company ↔ name
name ↔ bighit.